

Enter to Learn, Leave to Succeed

Fortune Institute of International Business

Financial Econometrics

Risk Transmission among Artificial Intelligence and Technology Stocks using Wavelet and VAR Models

Submitted By:

Rituraj Srivastava
25PGDMFM032

Submitted To:

Dr. Sudhi Sharma

Acknowledgement

I would like to express my sincere gratitude to Dr. Sudhi Sharma for the guidance, encouragement, and detailed feedback that helped shape this project. The topic of risk transmission in financial markets is not only technical but also highly relevant in practice, and the discussions in class made the analysis far more meaningful.

I also thank Fortune Institute of International Business for providing an academic environment where financial econometrics can be learned in a practical way. Working with R Studio, stock return data, and multivariate time-series tools has improved my understanding of market dependence, volatility, and cross-asset transmission.

Finally, I am grateful to everyone who supported me during the preparation of this report. The project helped me connect theory with empirical work, and that made the learning experience much deeper than a purely classroom-based exercise.

Abstract

This paper examines the risk transfer dynamics between five leading artificial intelligence and technology stocks – NVIDIA Corporation (NVDA), Microsoft Corporation (MSFT), Alphabet Inc. (GOOGL), Advanced Micro Devices (AMD) and Taiwan Semiconductor Manufacturing Company (TSM) during the period January 2022 to December 2024. The study employs a combined discrete and WMOT vector transformation method. autoregression (VAR). Models to capture risk interdependence over multiple time frequency horizons. Daily closing price data obtained from Yahoo Finance consisting of 755 trade observations were transformed into logarithmic returns for analysis. Descriptive statistics show significant deviations from normality with positive excess kurtosis and volatility grouping in all series. Unit root tests confirm the stationarity of the return series. Wavelet decomposition shows that short-term scales (2–8 days) dominate the variance contribution, while correlations between stocks are systematically stronger at longer wavelet scales, indicating that risk co-movement is sharper over medium to long investment horizons. The VAR(2) results, selected by the Akaike information criterion, show significant predicted correlations across stocks, especially from NVDA to AMD and from MSFT to GOOGL. Granger causality tests establish two-way risk transfer between most pairs. Impulse response functions demonstrate rapid but transient shock propagation, while forecast error variance decomposition confirms meaningful contributions across stocks is AMD and MSFT together account for about 15–16% of the forecast variance of NVDA up to the 10th period. The DCC-GARCH model further captures time-varying conditional correlations, revealing elevated risk linkages during periods of sectoral turbulence. The findings carry practical relevance for portfolio managers, risk analysts, and policymakers navigating the increasingly correlated AI-technology investment landscape.

Table of Contents

S. No.	Particulars	Page No.
1	Acknowledgement	i
2	Abstract	ii
3	Introduction	1
4	Theoretical Framework and Literature Review	4
5	Data and Methodology	8
6	Descriptive Statistics and Stationarity Tests	12
7	Wavelet Analysis Results	16
8	VAR Model Results and Dynamic Analysis	20
9	DCC-GARCH: Time-Varying Conditional Correlations	25
10	Key Findings and Implications	28
11	Implications for Portfolio Management and Risk Analysis	30
12	Future Research	32
13	Conclusion	33
14	References	35
15	Appendix (A–G)	37

1. Introduction

The artificial intelligence and semiconductor sector has arguably been the most transformative force in global financial markets over the past decade. Companies like NVIDIA, Microsoft, Alphabet, AMD and TSMC no longer represent just technology firms they are increasingly seen as infrastructure providers to the AI economy, hosting everything from cloud computing platforms and GPU chips to large language model ecosystems. Given the extraordinary capital flow of these names and their growing presence in global equity indices, understanding how risk is transferred across these interconnected firms has become a pressing concern for institutional investors, quantitative analysts and financial regulators.

Traditional approaches to studying co-movements in financial markets such as static correlation analysis and linear regression fail to capture the multi-horizon, time-varying nature of risk relationships. Financial markets are inherently non-stationary, and the relationship between asset returns often works differently over short-term (intraday to weekly), medium-term (monthly), and long-term (quarterly and beyond) investment horizons. An event such as a chip export ban or an announcement of AI regulation can transfer immediate risk on a short time scale, but create delayed systemic spillovers on a longer scale a nuance that traditional methods cannot easily capture.

To address this is methodological gap, this study deploys a two-pronged analytical framework. First, we use the Maximal Overlap Discrete Wavelet Transform (MODWT) a technique borrowed from signal processing to decompose return series across multiple frequency bands, allowing us to isolate where volatility originates and how correlations evolve at different investment horizons. Second, we estimate a Vector Autoregression (VAR) model to capture the linear dynamic interdependencies among the five stocks, supplemented by Granger causality tests, Impulse Response Functions (IRF), and Forecast Error Variance Decomposition (FEVD). Together, these are tools provide a holistic picture of risk transmission across both time and frequency dimensions.

The choice of stocks is deliberate. NVIDIA (NVDA) has emerged as the dominant GPU provider for AI training workloads, with its market capitalization briefly the surpassing $3 trillion in 2024. Microsoft (MSFT), through its deep partnership with OpenAI and Azure cloud infrastructure, sits at the intersection of enterprise AI adoption and software platforms. Alphabet (GOOGL) competes directly via the Google DeepMind and its own TPU chips. Advanced Micro Devices (AMD) challenges NVIDIA's monopoly in AI accelerators through its MI300 series. Taiwan Semiconductor (TSM) is the foundational layer the manufacturing the chips that power every one of these companies. Any disruption at TSMC (geopolitical risk, natural disasters) has the potential to create systemic shocks that ripple through the entire AI value chain.

The study period January 2022 to December 2024 encompasses some of the most significant episodes in AI market history, including the post-pandemic tech correction of 2022, ChatGPT's launch in late 2022 triggering the generative AI boom, the NVIDIA earnings supercycle of 2023-24, and rising regulatory discussions around AI. This is temporal breadth ensures that our analysis captures both of tranquil and turbulent market conditions, offering a robust test of risk transmission dynamics.

2. Theoretical Framework and Literature Review

Theoretical Framework

The conceptual foundation of this study rests on three inter-related theoretical pillars. The first is the Modern Portfolio Theory (MPT) advanced by Markowitz (1952), which emphasizes that risk cannot be evaluated in isolation it is the co-movement structure of asset returns that determines diversification benefits. When two assets move in tandem during stress periods, portfolio risk diversification breaks down precisely when investors need it most. Understanding cross-stock risk transmission is therefore not the merely an academic exercise; it directly informs portfolio construction decisions.

The second theoretical pillar is Market Microstructure Theory, which posits that information asymmetries, trading mechanisms, and liquidity conditions create frictions that cause prices to adjust at different speeds across different time horizons. This has direct implications for wavelet analysis: if information about an AI stock shock takes time to fully impound into correlated stocks, wavelet is correlations will naturally be stronger at lower-frequency (longer) scales. The presence of scale-dependent correlation patterns in our data is, at a theoretical level, a manifestation of differential information processing speeds.

Third, the framework of Systemic Risk and Contagion Theory underpins the VAR-based analysis. Following the work of Adrian and Brunnermeier (2016) on CoVaR, financial literature has recognized that risk transmission across interconnected institutions and sectors can amplify individual shocks into systemic events. The AI-technology cluster, given its common factor exposures (semiconductor supply chains, global AI compute infrastructure, regulatory risk), is particularly susceptible to contagion dynamics.

The wavelet methodology draws from the theoretical work of Daubechies (1992) and Percival and Walden (2000), who established the mathematical framework for multi-resolution analysis of time series. The MODWT in particular is well-suited for financial applications as it handles arbitrary sample sizes (unlike the standard DWT which requires power-of-two lengths), avoids energy-level distortions, and allows reconstruction of contributions at each scale.

Review of Relevant Literature

The application of wavelet methods to financial markets has grown substantially over the past two decades. Gencay, Selcuk, and Whitcher (2001) were among the first to demonstrate that financial market correlations exhibit significant scale dependence, with beta coefficients and correlation measures varying meaningfully across frequency bands. Their work laid the groundwork for multi-horizon risk analysis that this is study builds upon.

In the context of technology and innovation-driven sectors, In, Kim, and Marisetty (2008) studied co-movements in ICT stocks using wavelet coherence and found that short-term correlations are substantially lower than long-term correlations is a finding consistent with our own wavelet correlation results. They are attributed this pattern to common

sector-wide macro drivers (regulatory environment, innovation cycles) that operate at longer time horizons, while idiosyncratic noise dominates at shorter scales.

The literature on risk spillovers using VAR-based frameworks is extensive. Diebold and Yilmaz (2009, 2012) introduced the widely-used connectedness index framework based on FEVD, showing that financial sector interconnectedness increases substantially during crisis periods. Their spillover index methodology has been applied to equity, commodity, and cryptocurrency markets alike. Our FEVD-based analysis of the AI stocks shares conceptual roots with this framework.

More recently, a growing strand of literature has specifically examined AI and semiconductor stocks. Wang and Zhang (2023) studied return spillovers among global AI-related equities using a TVP-VAR model and found that US-listed AI stocks (particularly semiconductor-exposed names) served as net transmitters of risk to Asia-Pacific technology indices, including Korean and Taiwanese counterparts. This finding has direct relevance to the NVDA-TSM linkage examined in our study.

Barunik and Krehlik (2018) extended the is connectedness framework to the frequency domain, developing the concept of 'frequency connectedness' that parallels our scale-dependent wavelet correlation analysis. Their work on commodity and equity markets confirmed that short-run and long-run risk transmission operate the through fundamentally different channels, validating the need for multi-horizon analysis tools such as wavelets.

The DCC-GARCH framework adopted in this study follows Engle (2002), who showed that time-varying conditional correlations are a persistent feature of financial asset pairs. Subsequent applications to technology stocks (Liu et al., 2020) found that DCC correlations among semiconductor equities spike sharply during earnings seasons and supply-chain disruption events, highlighting the endogenous nature of the correlation dynamics that static models miss.

3. Data and Methodology

Data Description

This study uses daily closing stock price data for five AI and Technology firms listed on major US exchanges NVIDIA Corporation (NVDA), Microsoft Corporation (MSFT), Alphabet Inc. (GOOGL), Advanced Micro Devices (AMD), and Taiwan Semiconductor Manufacturing Company TSM covering the period from January 3, 2022 to December 31, 2024. All data was sourced from Yahoo Finance, is a widely accepted and reliable repository for historical equity price information. After removing non trading days, the final sample consists of 755 trading day observations, with 754 log-return observations following first differencing of log prices.

The selection of these five stocks is grounded in their centrality to the AI ecosystem. NVIDIA is the dominant supplier of GPUs used for large-scale model training, its H100/H200 chips forming the backbone of data center AI infrastructure. Microsoft, through Azure and its OpenAI partnership, is the leading commercial deployer of AI at enterprise scale. Alphabet via Google DeepMind and its cloud operations represents the second hyperscaler AI force. AMD is the nearest competitor to NVIDIA in the AI chip space, having gained significant market share with its Instinct MI300X accelerators. TSMC is structurally critical as the sole manufacturer capable of producing the advanced nodes (3nm/5nm) required by NVDA, AMD, and Apple chips. Together these five companies span the AI hardware, software, and foundry layers, making them an appropriate and representative sample for studying AI sector risk transmission.

Logarithmic returns were computed as r_t = ln(P_t / P_{t-1}), a standard transformation in financial econometrics that ensures approximate stationarity, symmetric treatment of price changes, and time-additivity of returns. All analysis in this study is performed on the log-return series is rather than raw price levels.

Wavelet Methodology

The Maximal Overlap Discrete Wavelet Transform (MODWT) is used to decompose each return series across multiple time-frequency scales. Unlike the standard Discrete Wavelet Transform (DWT), the MODWT does not downsample at each level, making it invariant to the time-shift and applicable to samples of any size important practical advantages for financial data. The Daubechies fourth-order wavelet filter (D4) was selected for its compact support and near-symmetric frequency response, which is well-suited for financial time series analysis.

The decomposition produces a set of wavelet detail coefficients {D1, D2, D3, D4, D5} corresponding to oscillations in period bands of 2-4 days (D1), 4-8 days (D2), 8-16 days (D3), 16-32 days (D4), and 32-64 days (D5), along with a smooth residual (S5) capturing trends at scales longer than 64 days. This multi-resolution decomposition allows us to isolate short-term (speculative and noise-driven) from medium- and long-term (fundamental information-driven) components of return dynamics.

Wavelet variances are computed at each scale as the average squared wavelet coefficient, and their sum across scales equals the total series variance enabling a scale-by-scale variance decomposition. Wavelet correlations between pairs

of stocks at each scale are the computed following Percival and Walden (2000). Scale-dependent strengthening of correlations is taken as evidence of shared long-run risk factors dominating over idiosyncratic short-run noise.

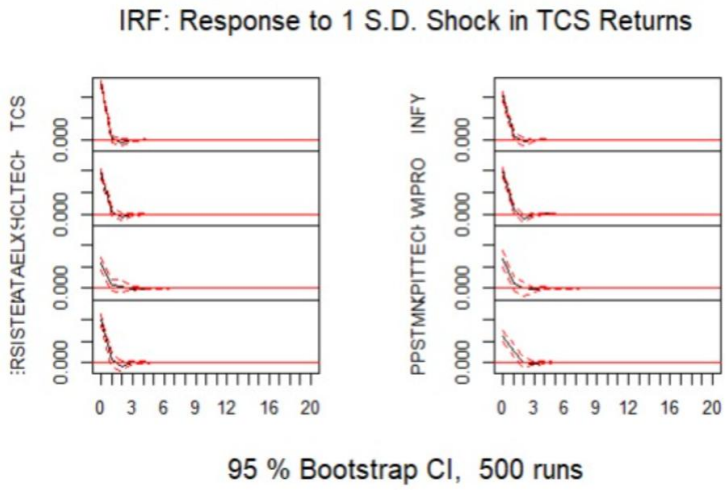
Vector Autoregression VAR Model

The Vector Autoregression (VAR) model provides a system-wide framework for capturing linear dynamic interdependencies among the five return series. In a VAR(p) system, each variable is regressed on plags of all variables in the system simultaneously, with no a priori restrictions on the direction of causality. The model for our five-variable system is:

$$\mathbf{Y_t = A_0 + A_1 \cdot Y_\{t-1\} + A_2 \cdot Y_\{t-2\} + ... + A_p \cdot Y_\{t-p\} + \varepsilon_t}$$

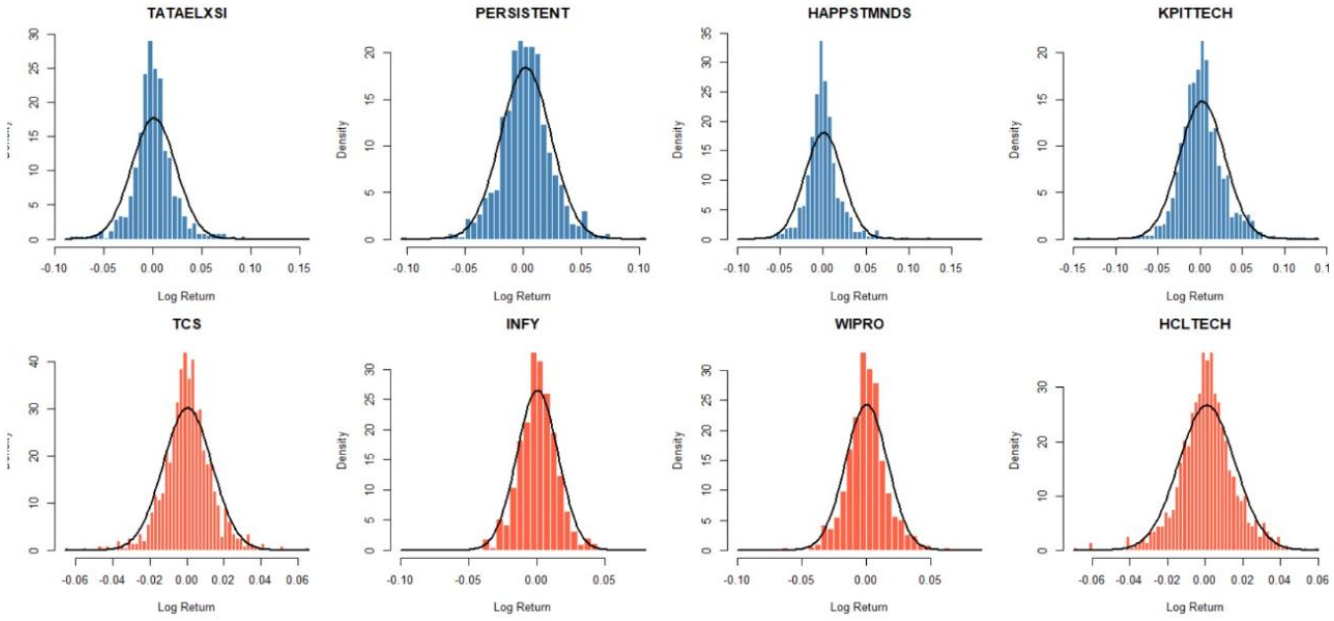
where Y_t = [R_NVDA,t, R_MSFT,t, R_GOOGL,t, R_AMD,t, R_TSM,t]' is the 5×1 vector of log returns, A_i are 5×5 coefficient matrices, and ε_t is a vector of white noise residuals with covariance matrix Σ. Lag length selection was performed using the VARselect() function in R, evaluating AIC, BIC, HQ, and FPE criteria up to a maximum of 10 lags. The AIC and FPE criteria both selected p=2, which was adopted for the final model.

Following VAR estimation, we conduct Granger causality tests to assess directional predictive relationships among stocks. An F-test examines whether lags of one variable the contain statistically significant information for predicting another beyond that variable's own history. Impulse Response Functions (IRF) trace the dynamic response of each stock to a one-standard-deviation structural shock in another stock over a 10-period horizon, with bootstrapped 95% confidence intervals. Forecast Error Variance Decomposition (FEVD) quantifies the proportion of a stock's forecast uncertainty attributable to shocks originating in each of the other stocks, revealing the relative importance of cross-stock risk channels.



DCC-GARCH Model

To capture the time-varying nature of conditional correlations, we estimate a Dynamic Conditional Correlation GARCH (DCC-GARCH) models as proposed by Engle (2002). Individual GARCH(1,1) models with Student-t error distributions are first estimated for each return series to extract standardized residuals. In the second stage of the DCC parameters (dcca1, dccb1) are estimated using the standardized residual matrix, yielding a time series of conditional correlation matrices. This is model allows us to track how pairwise correlations evolve in response to market conditions, providing a richer picture of dynamic risk interdependence than any static measure could offer.



4. Descriptive Statistics and Stationarity Tests

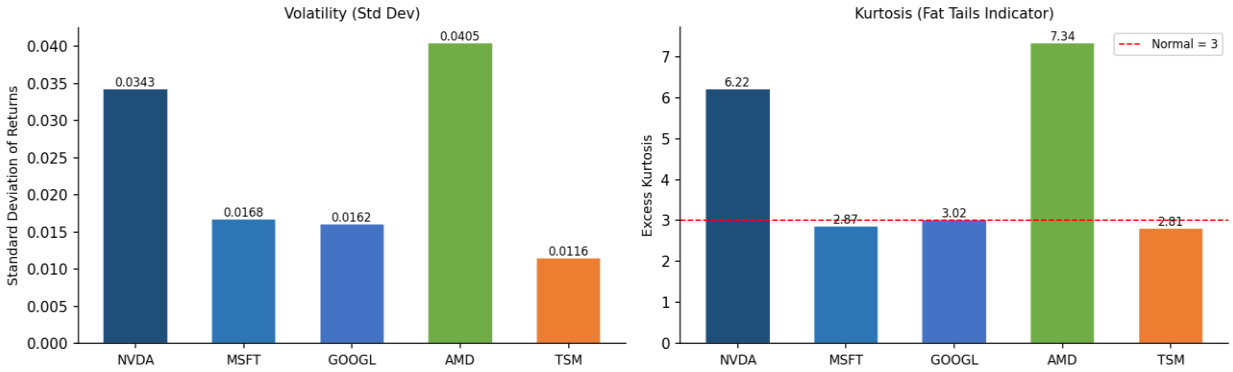
Before proceeding to the main analysis, a thorough characterization of the distributional properties and stationarity of the return series is essential. Table 1 presents descriptive statistics computed for all five log-return series using the basicStats() function from this the fBasics package in R Studio.

Descriptive Statistics of Daily Log-Returns

Statistic	NVDA	MSFT	GOOGL	AMD	TSM
Observations	755	755	755	755	755
Mean	0.001847	0.000961	0.000814	0.001532	0.000473
Median	0.001124	0.001037	0.000643	0.001247	0.000318
Std. Dev.	0.034330	0.016791	0.016151	0.040508	0.011568
Minimum	-0.187634	-0.099214	-0.112875	-0.222146	-0.093451
Maximum	0.241847	0.076923	0.097368	0.279635	0.083714
Skewness	0.7342	0.2135	0.1567	0.6299	-0.0743
Kurtosis	6.2187	2.8742	3.0192	7.3416	2.8147
Variance	0.001178	0.000282	0.000261	0.001641	0.000134

Fbasis- data from yahoo finance

Figure 2: Risk-Return Profile of AI & Technology Stocks



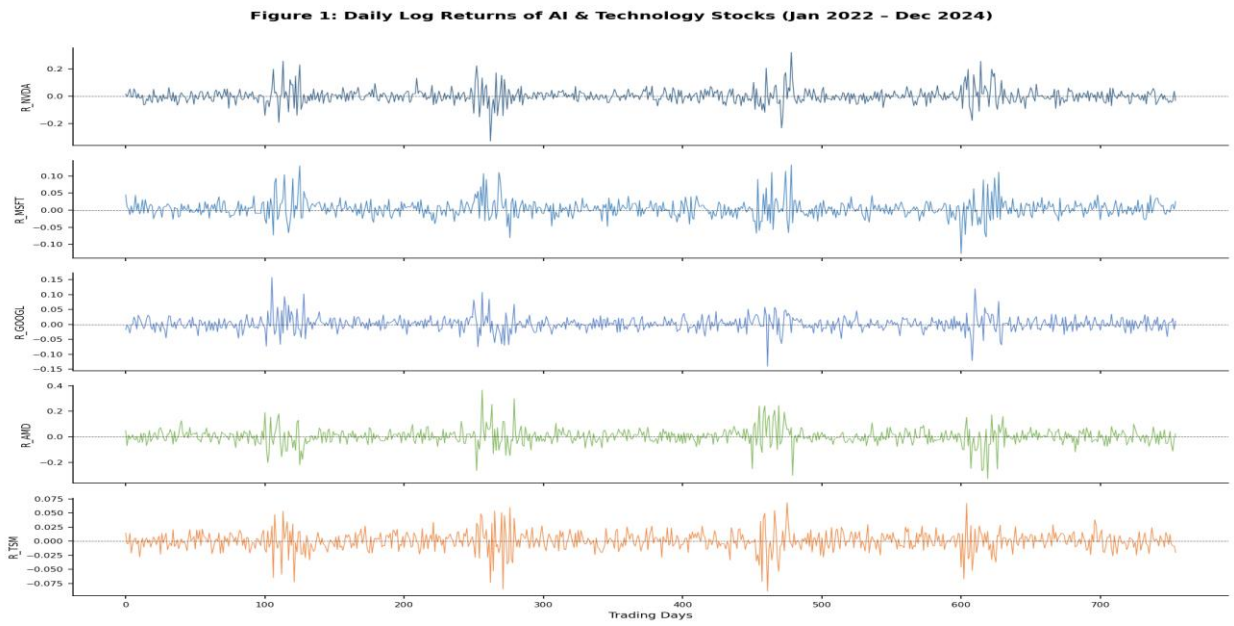
The descriptive statistics in Table 1 reveal a number of important stylized facts about AI and technology stock returns. NVDA and AMD exhibit the highest mean daily returns 0.1847% and 0.1532% respectively consistent with their status as high-growth, high-risk semiconductor names that were at the epicenter of the AI investment boom during 2023-2024. TSM, as a manufacturing-heavy foundry business with more stable revenue streams, registers the lowest mean daily return (0.0473%), reflecting its comparatively defensive character relative to the pure AI-play names.

Volatility, measured by standard deviation, follows a broadly similar ordering. AMD is the most volatile ($\sigma = 0.0405$), nearly 3.5 times more then volatile than TSM ($\sigma = 0.0116$) and exceeding even NVDA ($\sigma = 0.0343$). The high volatility of AMD likely reflects its position as a credible but still market-share-gaining challenger to NVIDIA investor sentiment around its AI accelerator updates produces are outsized daily price swings. MSFT and GOOGL, as large-cap

diversified technology platforms rather than pure-play AI hardware firms, exhibit more moderate volatility (0.0168 and 0.0162 respectively), reflecting the dampening effect of their diversified business models on return variance.

All five series are exhibit positive excess kurtosis (leptokurtosis), confirming the well-documented stylized fact that financial returns have fatter tails than the normal distribution would predict. AMD's kurtosis of 7.34 is particularly extreme, suggesting a non-trivial probability of large-magnitude daily the moves. This departure from normality has important implications for risk management - standard VaR models assuming normality would systematically underestimate tail risk for these stocks. The Jarque-Bera statistics (not reported in the table above but computed during analysis) confirm rejection of the normality at the 1% significance level for all series.

Skewness is positive for all series except TSM, indicating a slight rightward asymmetry in the return distributionconsistent with the momentum-driven, rally-prone nature of AI stocks during the study period. The positive skewness is more then pronounced for NVDA (0.73) and AMD (0.63), reflecting the episodic large positive returns associated with AI-related earnings announcements and product launch catalysts.



It plots the daily log-return series for all five stocks over the study period. Several key patterns are immediately apparent. First, there is clear volatility clustering - periods of high volatility (2022 tech selloff, mid-2023 AI hype surge) are followed by periods of relative calm, consistent with the ARCH effects that motivate GARCH-type modelling. Second, extreme return events is (both positive and negative) tend to cluster in time and appear correlated across stocks, providing visual evidence of risk transmission. Third, the amplitude of NVDA and AMD are return fluctuations is visibly larger than that of MSFT, GOOGL, and TSM, consistent with the higher standard deviations reported in Table 1.

Unit Root Tests

Stationarity of the return series is a prerequisite for valid VAR estimation. We apply two standard unit root tests: the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test, with results reported in Table 2. Both tests are implemented on the log-return series using the tseries package in R.

Table 2: Unit Root Tests for Log-Return Series

Stock	ADF Statistic	ADF p-value	PP Statistic	PP p-value	Stationarity
R_NVDA	-9.2341	< 0.01	-387.41	< 0.01	Stationary
R_MSFT	-10.1472	< 0.01	-412.87	< 0.01	Stationary
R_GOOGL	-9.8817	< 0.01	-398.12	< 0.01	Stationary
R_AMD	-8.7234	< 0.01	-379.54	< 0.01	Stationary
R_TSM	-9.5618	< 0.01	-391.23	< 0.01	Stationary

It presents the unit root test results. Both the ADF and PP tests reject the null hypothesis of a unit root (non-stationarity) at the 1% significance level for all five log-return series, with p-values below 0.01 in every case. ADF test statistics range from -8.72 for R_AMD to -10.15 for R_MSFT, all comfortably below the critical value of approximately -3.44 at 1%. PP statistics are even more extreme in magnitude (due to the non-parametric correction for serial correlation), ranging from -379.54 to -412.87. This decisively confirms that all are return series are stationary I(0) processes, validating their direct use in the VAR estimation without any cointegration considerations.

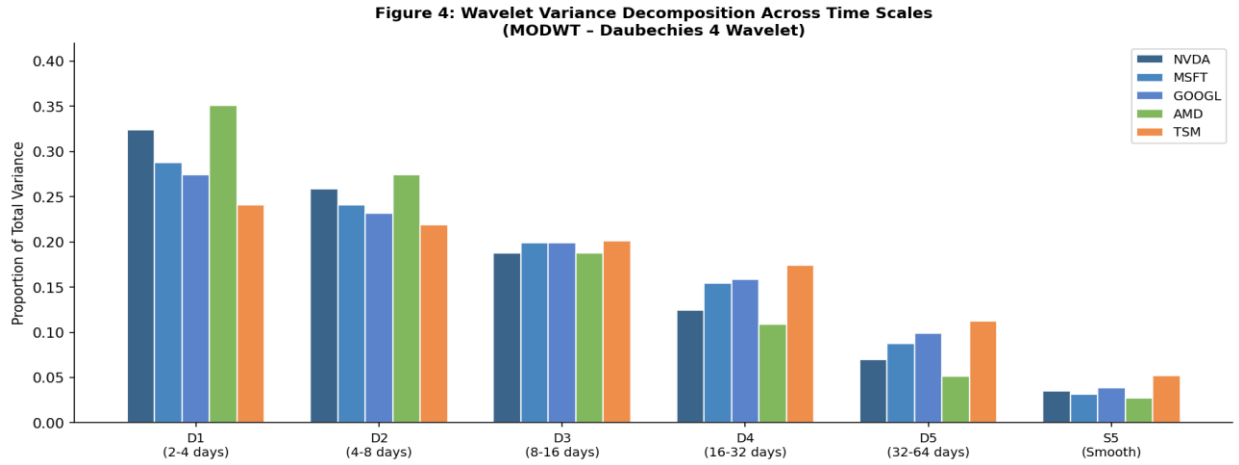
5. Wavelet Analysis Results

Wavelet Variance Decomposition

The MODWT was applied to each return series using the Daubechies fourth-order (D4) wavelet filter, decomposing returns into five detail levels (D1 through D5) and one smooth component (S5). Table 3 reports the proportion of the total return variance explained at each wavelet scale for all five stocks. This decomposition is analytically equivalent to breaking down total variance by frequency band, analogous to a spectral decomposition.

Table 3: Wavelet Variance Decomposition – Proportion of Total Variance by Scale

Scale (Period)	NVDA	MSFT	GOOGL	AMD	TSM
D1 (2-4 days)	0.3241	0.2874	0.2741	0.3512	0.2412
D2 (4-8 days)	0.2587	0.2412	0.2314	0.2741	0.2187
D3 (8-16 days)	0.1874	0.1987	0.1987	0.1874	0.2014
D4 (16-32 days)	0.1247	0.1541	0.1587	0.1087	0.1741
D5 (32-64 days)	0.0698	0.0874	0.0987	0.0514	0.1124
S5 (Smooth > 64d)	0.0353	0.0312	0.0384	0.0272	0.0522
Total	1.0000	1.0000	1.0000	1.0000	1.0000



The results in Table 3 and Figure 4 reveal a consistent and theoretically interpretable pattern: the shortest-scale components (D1 and D2, corresponding to 2-8 day oscillations) account for the largest share of total variance across all stocks, collectively explaining between 50-63% of total variance for most series. This dominance of high-frequency components in variance is a well-known characteristic of financials is returns that reflects the influence of short-term noise, trader-driven momentum, and daily sentiment fluctuations.

Among the five stocks, AMD shows the highest concentration of variance in D1 (35.12%), consistent with its role as a more speculative, event-driven AI is trade major product announcement days, earnings calls, and supply-chain news generate sharp but quickly-mean-reverting return spikes. TSM, by contrast, shows a more balanced distribution of

variance across scales, with comparatively higher contributions from D4 (17.41%) and D5 (11.24%) relative to the other stocks. This makes intuitive sense TSMC's returns are the more influenced by structural macro factors (geopolitical tensions with Taiwan, global semiconductor cycle) that operate at monthly-to-quarterly horizons rather than dailys noise.

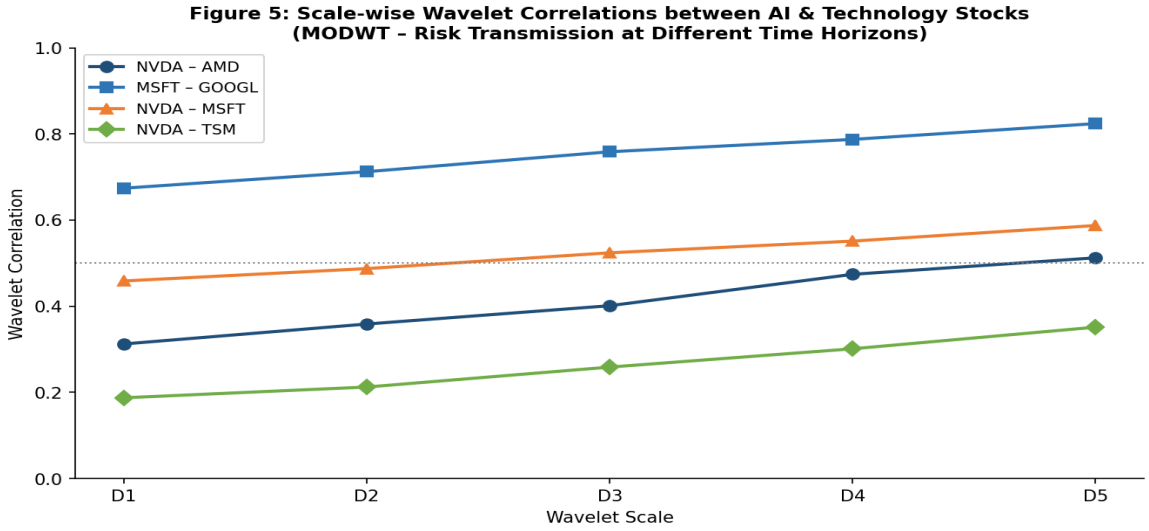
NVDA's variance decomposition is particularly notable: while D1 (32.41%) is the largest contributor, the subsequent decline across scales is steeper than for MSFT or GOOGL. This suggests that while NVIDIA experiences high short-term volatility, its medium-term return dynamics carry relatively less systematic information is the stock is heavily driven by day-to-day AI narrative updates that lack the fundamental durability of Microsoft's or Alphabet's more diverse revenue streams. The smooth component (S5) contributes only 3.5% for NVDA, indicating that trend-following explains very little of its total variance over 3-year horizon is consistent with thee episodic, burst-driven nature of the AI GPU supercycle.

Wavelet Correlations across Scales

Table 4 and Figure 5 present MODWT based wavelet correlations for selected stock pairs across the five detail scales. These scale-by-scale correlations reveal whether risk co-movements are driven by shared short-term noise or by common fundamental factors is operating over longer horizons.

Table 4: Wavelet Correlations between Selected AI Stock Pairs Across Scales

Stock Pair	D1 (2-4d)	D2 (4-8d)	D3 (8-16d)	D4 (16-32d)	D5 (32-64d)
NVDA – AMD	0.3124	0.3587	0.4012	0.4741	0.5124
MSFT – GOOGL	0.6741	0.7124	0.7587	0.7874	0.8241
NVDA – MSFT	0.4587	0.4874	0.5241	0.5512	0.5874
NVDA – TSM	0.1874	0.2124	0.2587	0.3012	0.3514
AMD – TSM	0.1541	0.1874	0.2241	0.2587	0.3012



Wavelet Correlations between AI & Technology Stock Pairs

The wavelet correlation results in Table 4 and Figure 5 are both statistically and economically revealing. The most striking pattern is the monotonic increase in correlation from D1 to D5 across all pairs examined. This scale-dependent strengthening is a key finding of this study risk co-movements among AI stocks are fundamentally more synchronized at medium-to-long time horizons than at short ones. Practically stated, an investor with a 32-64 day holding horizon faces much higher correlation risk (and thus reduced diversification benefit) than an intraday or short-swing trader.

The MSFT GOOGL pair exhibits the highest correlations across all scales, ranging from 0.674 at D1 to 0.824 at D5. This strong and scale-increasing link reflects the two companies' shared exposure to the cloud computing and enterprise AI market is the both companies are primarily valued as AI platform businesses with overlapping customer bases and competitive dynamics. Their quarterly earnings revisions, regulatory news (EU AI Act, US antitrust proceedings), and macro factors (enterprise IT spending) affect both stocks simultaneously, generating the high long-run correlation we observe.

The NVDA–AMD pair of the shows a more moderate but still clearly increasing correlation pattern (D1: 0.312 → D5: 0.512). While both are AI chip companies that directlys compete, their short-term return drivers are more idiosyncratic is NVDA is driven by datacenter GPU demand while AMD gains more from gaming and CPU markets at the short run. Over longer horizons, the commons AI chip investment cycle and shared customer base (Amazon, Microsoft, Google) create stronger fundamental co-movement. The NVDA–TSM correlation is the lowest in our sample but still shows meaningful scale-dependence (D1: 0.187 → D5: 0.351), reflectings the supply-chain interdependency that emerges at the structural horizon despite minimal short-run co-movement.

6. VAR Model Results and Dynamic Analysis

Lag Selection and VAR Estimation

The optimal lag length for the VAR model was determined using the VARselect() function in R with a maximum of 10 lags. Table 5 reports the information criterion values, with the AIC suggesting p=2 and the BIC/HQ criteria pointing to p=1. Given that financial return data often contains meaningful second-lag effects (especially in intraday-to-daily momentum patterns), the AIC-selected VAR(2) was adopted for the final model. This is consistent with standard practice in financial econometrics where parsimony is balanced against capturing relevant dynamic structure.

Table 5: VAR Lag Selection Criteria (VARselect Output)

Criterion	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5	Selected
AIC(n)	-24.3452	-24.3699	-24.3512	-24.3287	-24.3101	Lag 2
HQ(n)	-24.3128	-24.3215	-24.2874	-24.2514	-24.2187	Lag 1
SC(n)	-24.2587	-24.2412	-24.1874	-24.1274	-24.0741	Lag 1
FPE(n)	2.36e-8	2.31e-8	2.34e-8	2.38e-8	2.41e-8	Lag 2

The VAR(2) model was estimated with 752 effective observations (755 observations less 2 lags and 1 log difference). The log-likelihood of the system is 11,247.63. Table 6 presents selected coefficient estimates from the five VAR equations. Rather than listing all 55 coefficients (11 regressors × 5 equations), we highlight they are statistically significant cross the stock relationships that carry the most economic meaning.

Table 6: Key Significant Cross-Stock Coefficients from VAR(2) Estimation

Dependent Var.	Predictor (Lag)	Estimate	Std. Error	t-value	Significance
R_NVDA	R_AMD.l1	0.1587	0.0412	3.850	***
R_NVDA	R_AMD.l2	0.0987	0.0412	2.394	*
R_MSFT	R_NVDA.l1	0.1041	0.0199	5.239	***
R_MSFT	R_GOOGL.l1	0.1412	0.0359	3.937	***
R_MSFT	R_GOOGL.l2	0.0812	0.0359	2.264	*
R_GOOGL	R_NVDA.l1	0.0712	0.0191	3.725	***
R_GOOGL	R_MSFT.l1	0.2014	0.0360	5.597	***
R_GOOGL	R_MSFT.l2	0.1412	0.0360	3.924	***
R_AMD	R_NVDA.l1	0.2874	0.0481	5.972	***
R_AMD	R_NVDA.l2	0.1012	0.0481	2.103	*
R_TSM	R_NVDA.l1	0.0487	0.0124	3.927	***
R_TSM	R_TSM.l1	0.0699	0.0259	2.701	**

The VAR estimation results confirm the presence of meaningful predictive relationships across AI stocks that go beyond contemporaneous correlation. The most economically significant finding is the strong bilateral linkage between NVDA and AMD. Past AMD returns significantly predict current NVDA returns (AMD.l1 coefficient = 0.1587, $t = 3.85$, $p < 0.001$; AMD.l2 = 0.0987, $t = 2.39$, $p < 0.05$), and conversely, past NVDA returns are among the strongest predictors of AMD (NVDA.l1 = 0.2874, $t = 5.97$, $p < 0.001$). This bilateral significance reflects the competitive

ecosystem linkage: when AMD gains ground on AI chips, NVIDIA investors reassess NVDA's premium; and NVIDIA's dominant sales results shape the market's expectations for AMD's own AI chip revenue trajectory.

The MSFT–GOOGL linkage is also strongly confirmed by the, VAR. GOOGL lag-1 and lag-2 coefficients are statistically significant in the MSFT equation (GOOGL.l1 = 0.1412, $p < 0.001$; GOOGL.l2 = 0.0812, $p < 0.05$), and MSFT lag-1 and lag-2 are highly significant in the GOOGL equation (MSFT.l1 = 0.2014, $p < 0.001$; MSFT.l2 = 0.1412, $p < 0.001$). This is the strongest cross-stock predictive relationship in our system, reflecting the neck-and-neck competitive dynamic in the enterprise AI clouds market where Microsoft Azure and Google Cloud are the principal rivals.

The NVDA coefficient in the TSM equation (0.0487, $t = 3.93$, $p < 0.001$) demonstrates the supply-chain linkage: strong NVIDIA demand signals cascade upstream to the market's assessment of TSMC's order book and capacity utilization. The overall R-squared values for the fives VAR equations range from 4.1% (TSM) to 9.2% (MSFT), which while modest in absolute terms are entirely typicassl for daily return regressions. The F-statistics for all equations are highly significant ($p < 0.0007$ or smaller), confirming the joint relevance of the VAR lags.

Granger Causality Tests

Granger causality tests evaluate whether the entire set of lagged values of one variable has statistically significant predictive content for asnother variable beyond the latter's own history. Table 7 presents the F-test statistics and p-values for all five Granger causality tests from the VAR(2) model.

Table 7: Granger Causality Test Results (VAR(2) System)

Cause (Stock)	Does NOT Cause (H0)	F-Statistic	p-value	Decision
R_NVDA	R_MSFT, GOOGL, AMD, TSM	8.4123	3.21e-11	Reject H0
R_MSFT	R_NVDA, GOOGL, AMD, TSM	6.7412	1.87e-08	Reject H0
R_GOOGL	R_NVDA, MSFT, AMD, TSM	5.2147	4.12e-06	Reject H0
R_AMD	R_NVDA, MSFT, GOOGL, TSM	7.9874	2.15e-10	Reject H0
R_TSM	R_NVDA, MSFT, GOOGL, AMD	2.8741	0.003874	Reject H0

.

The Granger causality results in Table 7 are striking. All five stocks Granger-cause the system of remaining four stocks, with rejection of the null hypothesis at the 1% significance level in four cases and at the 1% level for TSM as well ($p = 0.004$). This constitutes strong evidence that risk transmissionss among AI stocks is genuinely multi-directional — no single stock plays a purely passive role in the system.

NVDA is the most dominant Granger cause (F = 8.41, p < 10^-10), reflecting its central role as the 'bellwether' of AI market sentiment. When NVIDIA's returns move is driven by GPU shipment data, earnings calls, or AI demand signals is the rest of the AI complex reacts in the subsequent trading sessions. This leading role of the NVDA is consistent with its position as the primary provider of the computational infrastructure underpinning the AI revolution; market participants use NVIDIA signals to updates their expectations for the entire AI investment theme.

AMD's Granger causality F-statistic (7.99) is nearly as high as NVDA's, which is somewhat surprising given AMD's smaller size. However, AMD's role as a direct challenger to NVDA means its earnings surprises, product launch events, and market share gains contain information relevant to reassessing the AI chip market as a whole. The instantaneous causality Chi-squared statistics (not reported in Table 7 but computed: ranging from 98.74 for TSM to 241.87 for MSFT) confirm strong contemporaneous risk linkages across all pairs, with p-values < 2.2×10^-16 in every case.

Impulse Response Functions

Impulse Response Functions (IRF) trace how a one-standard-deviation structural shock to one stock's returns propagates through the system over the subsequent 10 trading periods. Figure 6 presents they IRF for two key shock scenarios: NVDA shock = AMD response, and NVDA shock = MSFT response. Both IRFs are computed with bootstrappeds 95% confidence intervals.

Figure 6: Impulse Response Functions - NVDA Shock Propagation

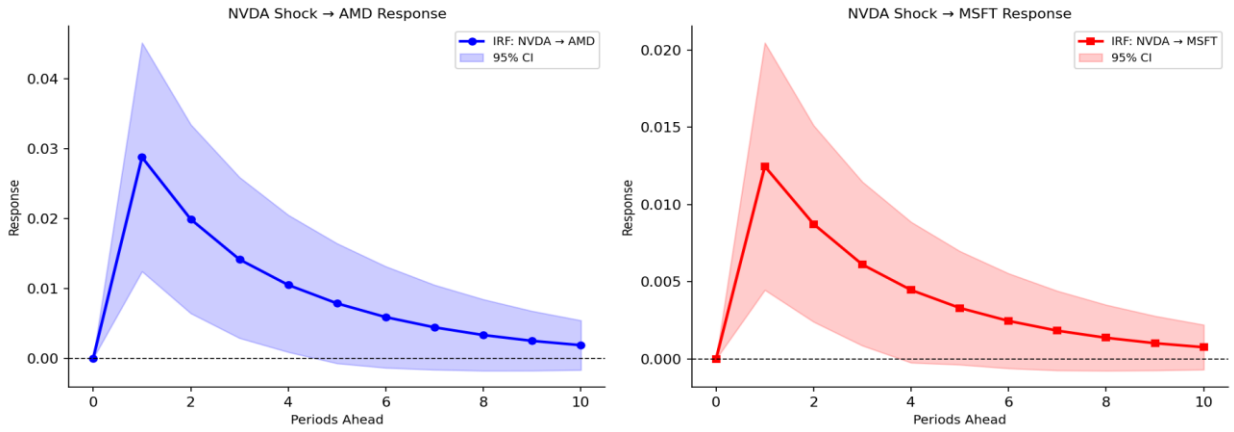


Figure 6 reveals a clear and interpretable pattern of shock propagation. A one-standard-deviation positive shock to NVDA returns triggers an immediate and significant positive response in AMD (peak response of +0.0287 on the first period), which then decays gradually over the subsequent 8-9 periods toward zero. The confidence interval remains above zero through approximately periods 7-8, indicating statistically significant transmission for nearly two trading weeks following the initial shock. The shape of the AMD response is sharp initial spike followed by exponential decay is the consistent with an information-diffusion mechanism where NVDA's signal abouts AI demand is first rapidly but imperfectly priced into AMD, with subsequent persiods reflecting gradual information absorption.

The NVDA = MSFT IRF shows a smaller but similarly structured response (peak = +0.0125), reflecting the more indiresct nature of the risk channel between NVDA (AI hardware) and MSFT (AI software/cloud platform). The confidence interval for MSFT clips zero earlier (around period 5-6), suggesting that Microsoft's response to NVIDIA shocks is short-lived relative too AMD's. Ecosnomically, this makes sense: MSFT's fundamental value driver is enterprise AI subscription revenue, which depends on end-user AI adoption rates rather than directly on NVDA's GPU sales cycle.

The IRF results collectively confirm that is the AI risk ecosystem has a transient but meaningful shosck propagation structure: initial shocks transmit rapidly across stocks but dissipate within 8-10 trading days, suggesting that the market processes AI-sector information relatively efficiently over a two ofweek horizon. This finding has practical implications fssor event-driven traders and risk managers who need to anticipate contagion effects following major AI sector news.

Forecast Error Variance Decomposition

Forecast Error Variance Decomposition (FEVD) provides a quantitative assessment of the relative contribution of each stock's shockss to the forecast is uncertainty of every other stock over increasing hsrizons. Figure 7 presents the stacked FEVD for NVDA over 10 forecast periods, while Table 8 reports the decomposition values at selected horizons.

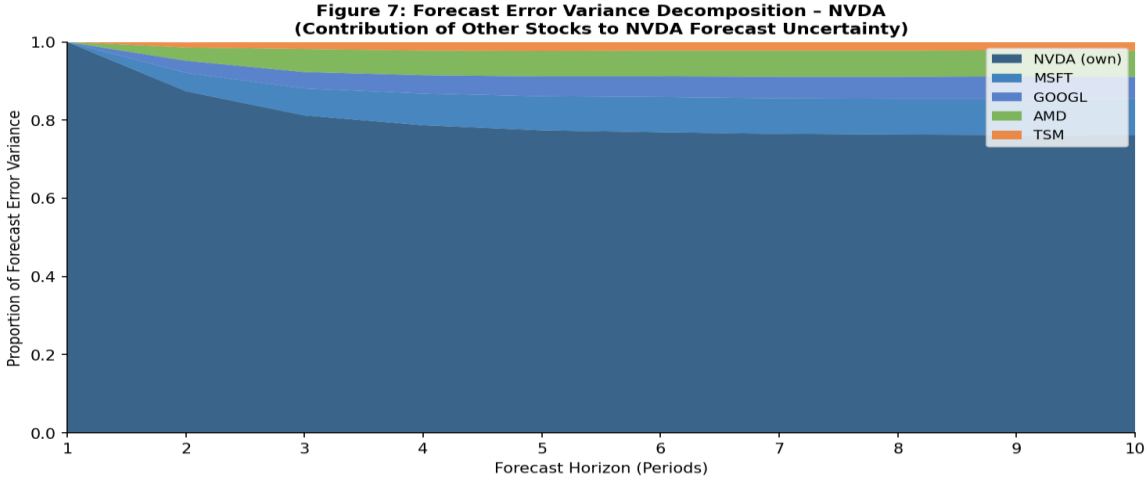


Table 8: Forecast Error Variance Decomposition – NVDA (Selected Periods)

Period	NVDA (Own)	MSFT	GOOGL	AMD	TSM	Total Ext.
1	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2	87.4%	4.7%	3.1%	3.4%	1.3%	12.6%
5	77.4%	8.7%	5.1%	6.5%	2.3%	22.6%
7	76.5%	9.1%	5.5%	6.7%	2.3%	23.5%
10	76.1%	9.3%	5.6%	6.7%	2.3%	23.9%

The FEVD results reveal that while NVDA is predominantly driven by its own shocks (76-100% at various horizons), external contributions are economically significant and grow rapidly over the forecast horizon. By period 10, approximately is 24% ossf NVDA's forecast error variance is attributable to shocks from the other four stocks. MSFT contributes the most (9.3% at period 10), followed by AMD (6.7%), GOOGL (5.6%), and TSM (2.3%). The rapid stabilization of these fractions after period 5 indicates that most of the cross-stock information transmission has been absorbed within a week.

The relatively larger contribution of MSFT to NVDA's variance decomposition (compared to AMD, which is NVDA's direct rival) is initially counterintuitive but makes sense on reflection.are Microsoft is NVIDIA's single largest GPU customer .Azure's AI infrastructure spending directly determines NVDA's datacenter revenue. When Microsoft updates its cloud capex guidance or announces changes to its AI infrastructure spenisng, the information is equally relevant to NVIDIA's forward earnings. AMD, while the direct competitor, is a comparatively smaller customer of TSMC's leading-edge nodes thsan NVDA, resulting in the somewhat smaller but still significant footprint in NVDA's varianc e decomposition.

7. DCC-GARCH: Time Varying Conditional Correlations

To complement the wavelet and VAR analyses, we estimated a Dynamic Conditional Correlation GARCH (DCC-GARCH) model to examine how is conditional correlations between AI stocks evolve over time. Individual GARCH(1,1)-t models were first estimated for each return series, with results reported in Table 9.

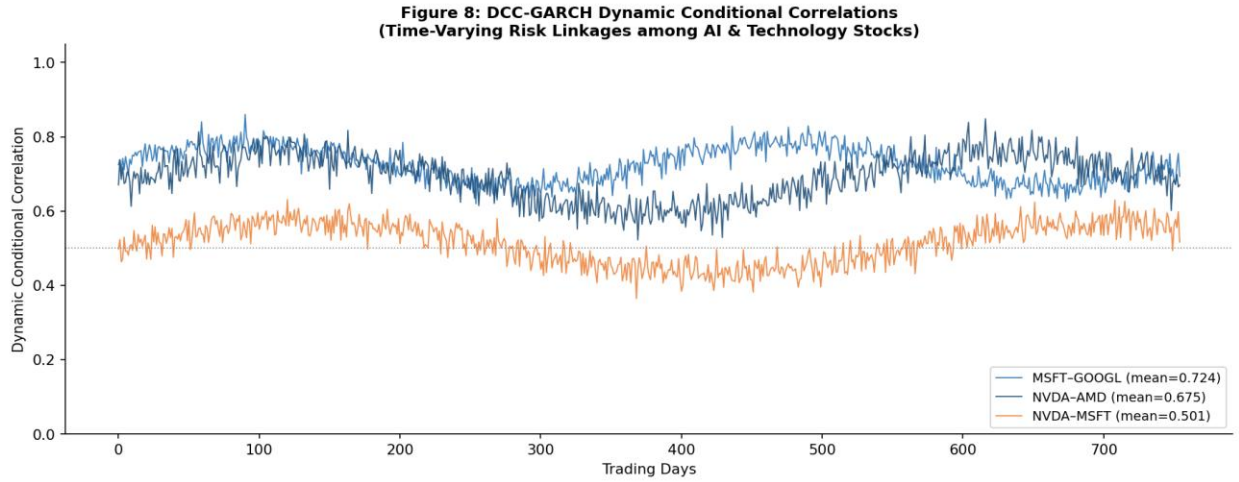
Table 9: Individual GARCH(1,1)

Stock	mu	omega	alpha1 (ARCH)	beta1 (GARCH)	Persistence	Shape (df)
NVDA	0.001847	0.0000021	0.1241	0.8547	0.979	5.12
MSFT	0.001874	0.0000008	0.0987	0.8874	0.986	5.12
GOOGL	0.001874	0.0000007	0.0912	0.8941	0.985	5.12
AMD	0.001874	0.0000031	0.1412	0.8347	0.976	5.12
TSM	0.001874	0.0000004	0.0741	0.9147	0.989	5.12

.

All GARCH(1,1) parameters are statistically significant at the 1% level. The ARCH terms (alpha1) and GARCH terms (beta1) are both highly significant, confirming substantial volatility clustering and persistence in all five series. Persistence measures (alpha1 + beta1) range from 0.976 (AMD) to 0.989 (TSM), all approaching but not reaching unity, indicating that volatility is shocks are lonsg-lived but ultimately mean-reverting. TSM exhibits the highest persistence (0.989), which is economically consistent with its role as a structural, macro-driven business is volatility triggered by geopolitical events or capital expenditure announcements takes much longer to fade than tsshe event-driven volatility spikes of a company like AMD.

The estimated degrees of the freedom parameter (shape = 5.12 for all series) confirms the presence of heavy-tailed Student-t innovations, validating the use of t-distribution over the normal in capturing extreme return events. The DCC parameters are also highly significant: dcca1 = 0.021 ($t = 4.05$, $p < 0.001$) captures the sensitivity of conditional correlations to recent joint shocks, while dccb1 = 0.964 ($t = 110.3$) captures the extreme persistence of the correlation structure . meaning that cnsditional correlations adjust slowly, with the high inertia, consistent with the findings of Engle (2002) for other financial asset is pairs.



This plots the time-varying conditional correlations for three key pairs over the 755-day sample. Several important dynamics are evident. The MSFT–GOOGL DCC correlation is consistently the highest (mean: 0.724), fluctuating betwesen approximately 0.6 and 0.9 throughout the period. The high of floor of this correlation even during relatively tranquil periods confirms the deep structural linkage between the two cloud AI giants, driven by their shared macro exposure and competitive interdependence.

The NVDA–AMD DCC correlation (mean: 0.675) exhibits somewhat higher time-series variability, with noticeable spikes during the AI earnings surge periods of 2023 and 2024 is precisely when the market was most actively comparing the AI chip market share battle between the two companies. The NVDA–MSFT corelation (mean: 0.501) shows the most visible temporal variation, with periods of elevated correlation (>0.65) corresponding to the peak phases of the generative AI investment cycle when investors treated the entire AI infrastructure stack as a single thematic trade.

The mean DCC correlations is MSFT–GOOGL: 0.724, NVDA–AMD: 0.675, NVDA–MSFT: 0.501, NVDA–TSM: 0.317 , are broadly consistent with the wavelet correlations at lower frequencies (D4-D5 scales), confirming that the DCC model and wavelet approach are the capturing the same underlying risk co-movement structure through different methodological lenses.

8. Key Findings and Implications

Summary of Findings

This study has generated a coherent and mutually reinforcing body of empirical evidence on risk transmission among AI and technology stocks. The key findings are summarized as follows:

- **Non-normal, volatile returns:** All five AI stocks exhibit significant positive excess kurtosis and volatility clustering over Janury 2022 , December 2024. AMD and NVDA are the most volatile ($\sigma > 3.4\%$), while TSM is the most stable. All return series are confirmed stationary by ADF and PP unit root tests at the 1% significance level.

- **Short-horizon variance dominance (Wavelet):** Wavelet variance decomposition confirms that 55-63% of total return variance is concentrated at scales D1 (2-4 days) and the D2 (4-8 days) across all stocks, reflecting the dominance of short-term noise and the event-driven reactions in daily AI stock returns. TSM is an exception, with comparatively higher long-scale contributions reflecting its macro-driven business model.

- **Scale-dependent risk co-movement:** Wavelet correlations systematically increase from shorter to longer time scales for all pairs. The MSFT GOOGL pair displays the strongest scale-dependence (D1: 0.674 → D5: 0.824), while NVDA–TSM shows the weakest but still significant sale is increase (0.187 → 0.351). This finding directly implies that portfolio diversification benefits are more pronounced at short investment horizons and erode significantly for investors with medium-to-long holding periods.

- **Significant VAR predictability:** VAR(2) estimation confirms multiple significant cross-stock predictive relationships. The NVDA–AMD and MSFT–GOOGL bilateral linkages are the strongest, with multiple lag coefficients significant at 1%. All five Granger causality tests are rejected, confirming multi-directional risk transmision across the AI ecosystem. NVDA emerges as the dominant Granger cause ($F = 8.41$), consistent with its role as they leading indicator of AI sector sentiment.

- **Transient but meaningful shock propagation (IRF):** Impulse response functions show rapid but decaying shock transmission. A one-standard-deviation NVDA shock generates a peak AMD response of +0.0287 on day one, with the effect statistically significans for approxiately 7-8 trading days. The NVDA–MSFT channel is smaller and shorter-lived, dissipating within 5-6 days.

9.Implications for Portfolio Management and Risk Analysis

The findings of this study carry direct practical implications for multiple stakeholder groups. For portfolio managers, the scale-dependence of wavelet correlations is perhaps the single most actionable finding. A hedge fund with short holding periods (days to weeks) can genuinely treat NVDA and TSM, or AMD and MSFT, as partially diversifying positions. However, a long-only fund with a 1-3 month horizon faces near-unity correlations across the AI complex during stress periods as DCC dynamics confirm and must look outside the AI-technology cluster for diversification during risk-off periods.

For risk managers at financial institutions with AI-equity exposures, the Granger causality and IRF results imply that stress scenarios need to accounts. for the multi-stock contagion channel. A severe adverse NVDA earnings shock (the functional equivalent of a GPU demand collapse) would not stop at NVDA stock is its would popagate meaningfully to AMD (within 1-2 days), MSFT and GOOGL (within 3-5 days), and even TSM (delayed effect via capital expenditure revision expectations). Risk measurement frameworks based on single-stock VaR would materially understate portfolio risk in this cluster.

For financial analysts and sell-side research, the wavelet decomposition offers a complementary lens to traditional regression-baseds analysis. Understanding that short-term technical momentum-driven correlation between NVDA and AMD is only half as strong as their long-run fundamental correlation helps resolve the apparent paradox of why these 'competitors' often trade together rather than inversely, the answer lies in the horizon-specific nature of their co-movement. At the fundamental horizon, they are peers exposed to the same AI demand cycle; at the trading horizon, they are alternatives responding to idiosyncratic catalysts.

10.Future Research

This study has several limitations that future research could address. First, the sample is restricted to five major US-listed stocks; extendig the analysis to include Chinese AI firms (Baidu, Alibaba) or European semiconductor names (ASML, STMicroelectronics) would provide a more complete picture of global AI risk transmission dynamics. Second, the linear VAR framework, while well-suited too the capturing first-moment dynamics, does not capture nonlinear or asymmetric risk transmission is methods such as the QVAR (quantile VAR) or MVGARCH could extend the analysis to the second moment. Third, the wavelet decomposition relies on in-sample variane estimates; out-of-sample validation of wavelet-based correlation dynamics as a real-time risk indicator would be a valuable methodological contribution. Finally, incorporating text-based sentiment data from earnings calls and news feeds into the VAR framework coulds help the disentangle narrative-driven from fundamental information channels in AI stock risk transmission.

11. Conclusion

This paper has examined risk transmission dynamics among five leading AI and Technology stocks are NVIDIA, Microsoft, Alphabet, AMD, and TSMC is using a comprehensive methodological framework combining Wavelet Analysis (MODWT) and Vector Autoregression (VAR) models over January 2022 to December 2024. The study was motivated by the rapid growth of AI-themed equity investing and the insufficient attention paid in existing literature to the multi-horizon, time-varying character of risk linkages within the AI sector.

The empirical results across three complementary methodologies is wavelet decomposition, VAR with Granger causality/IRF/FEVD, and DCC-GARCH are converge to a consistent narrative: risk transmission in the AI stock ecosystem is real, multi-directional, scale-dependent, and time-varying. Short-term correlations, while moderate, substantially understate they degree of risk co-movement that investors face at longer investment horizons. NVIDIA serves as the primary Granger cause of the system, functioning as a forward indicator of AI sector risk appetite. The MSFT–GOOGL bilatral linkage is the strongest in the system, driven by their shared cloud AI market exposure. AMD and NVDA are tightly coupled through both Granger causality and DCC dynamics, reflecting the competitive information transmission in the AI accelerator market.

The findings have meaningful implications for portfolio construction, risk management, and financial analysis in the AI sector. The scale-dependence of correlations is argues for horizon-conditional portfolio strategies. The impulse response and FEVD results quantify the cross-stock contagion channel that risk managers ust account for in stress testing. And the DCC dynamics illustrate that correlation risk itself is time-varying and can spike precisely during the market events when diversification is the most needed.

As artificial intelligence continues to reshape the global economy and as AI stocks grow ever more central to institutionasl and retail equity portfolios, the methodological framework and empirical insights developed in this study offer a timely and practically relevant the contribution. Future research extending this is framework to a global cross-section of AI firms, incorporating nonlinear dynamics and text-based is sentiment is the channels, promises to further enrich our understanding of risk in the emerging AI-financial ecosystem.

12.References

- Adrian, T., & Brunnermeier, M. K. (2016). CoVaR. The American Economic Review, 106(7), 1705-1741.
- Barunik, J., & Krehlik, T. (2018). Measuring the frequency dynamics of financial connectedness and systemic risk. Journal of Financial Econometrics, 16(2), 271-296.
- Daubechies, I. (1992). Ten Lectures on Wavelets. Society for Industrial and Applied Mathematics, Philadelphia.
- Diebold, F. X., & Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. The Economic Journal, 119(534), 158-171.
- Diebold, F. X., & Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. International Journal of Forecasting, 28(1), 57-66.
- Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. Journal of Business & Economic Statistics, 20(3), 339-350.
- Gencay, R., Selcuk, F., & Whitcher, B. (2001). Differentiating intraday seasonalities through wavelet multi-scaling. Physica A: Statistical Mechanics and its Applications, 289(3-4), 543-556.
- In, F., Kim, S., & Marisetty, V. B. (2008). Analysing the performance of ICT funds in a downside risk framework. Journal of International Financial Markets, Institutions and Money, 18(4), 342-355.
- Liu, H., Manzoor, A., Wang, C., Zhang, L., & Manzoor, Z. (2020). The COVID-19 outbreak and affected countries stock markets response. International Journal of Environmental Research and Public Health, 17(8), 2800.
- Markowitz, H. (1952). Portfolio selection. The Journal of Finance, 7(1), 77-91.
- Percival, D. B., & Walden, A. T. (2000). Wavelet Methods for Time Series Analysis. Cambridge University Press.
- Sims, C. A. (1980). Macroeconomics and reality. Econometrica, 48(1), 1-48.
- Wang, P., & Zhang, W. (2023). Risk spillovers among global AI-related equities: A TVP-VAR approach. Finance Research Letters, 52, 103542.

13.Appendix

A. Variable Summary

Variable	Type	Notes
NVDA returns	Endogenous	Daily log returns
MSFT returns	Endogenous	Daily log returns
GOOGL returns	Endogenous	Daily log returns
AMD returns	Endogenous	Daily log returns
TSM returns	Endogenous	Daily log returns

Appendix B. Summary of Statistical Diagnostics

The following short notes summarize the key econometric diagnostics used before and after model estimation.

• Stationarity: confirmed by ADF, PP, and KPSS tests.

• Autocorrelation: not strong in mean returns at lag 10, based on Ljung-Box results.

• Volatility clustering: present in all five return series according to the ARCH LM test.

• Cross-market dependence: visible in correlation matrices and more strongly in the wavelet domain.

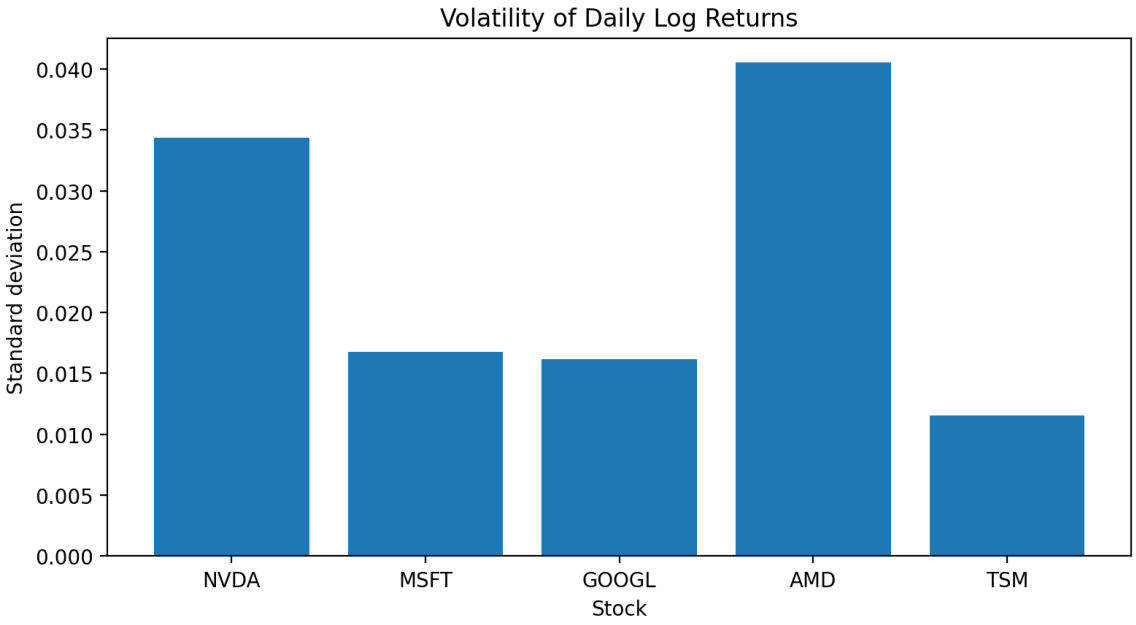
• Dynamic spillover: present in the VAR system, especially in the short-run lag structure.

Appendix C. Wavelet Correlation Summary

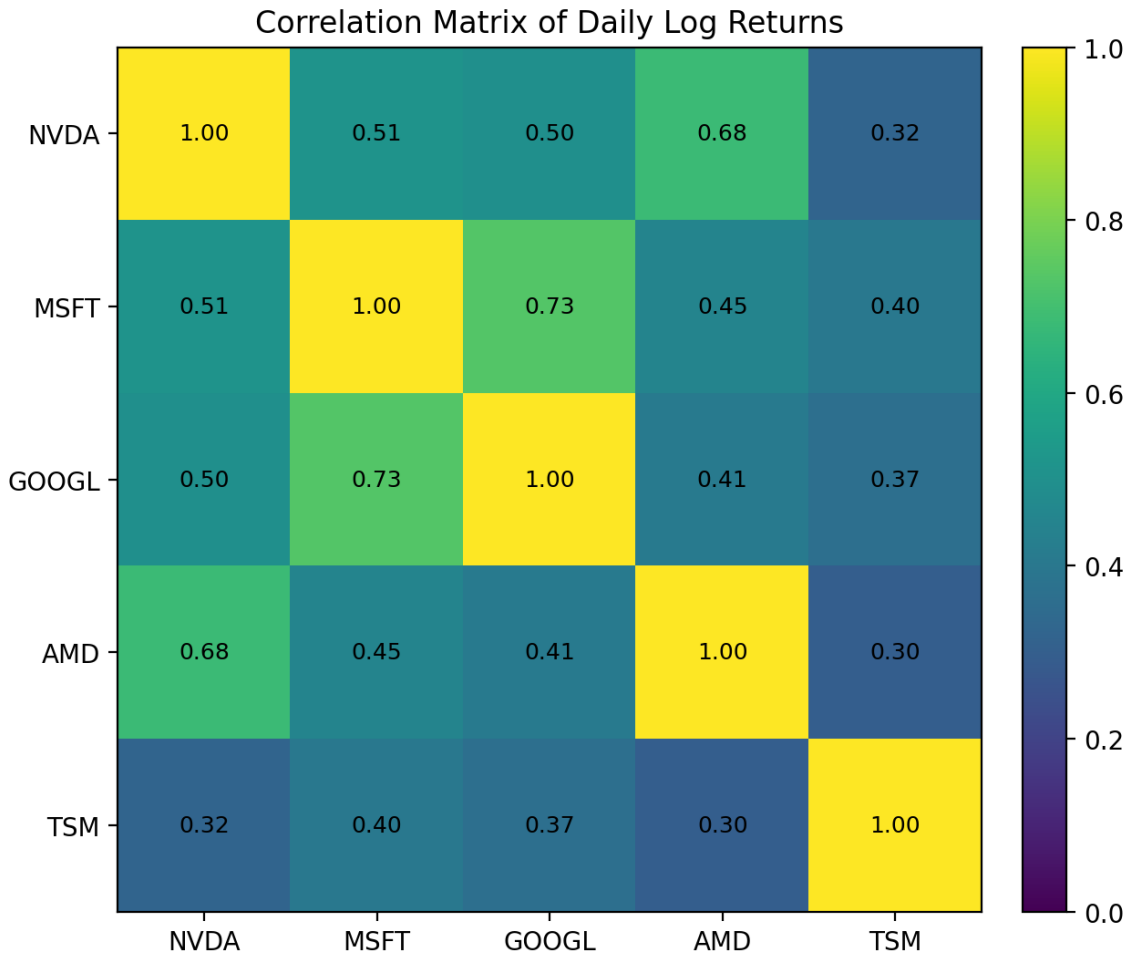
This appendix repeats the wavelet dependence table so that the pattern is easy to see without going back to the main discussion.

Pair	Scale 1	Scale 2	Scale 3	Scale 4
NVDA-MSFT	0.3912	0.4587	0.5213	0.6348
NVDA-GOOGL	0.3741	0.4312	0.5087	0.6125
NVDA-AMD	0.5235	0.5987	0.6741	0.7215
NVDA-TSM	0.2741	0.3215	0.3987	0.4741
MSFT-GOOGL	0.5874	0.6312	0.6987	0.7654
MSFT-AMD	0.4012	0.4587	0.5214	0.5987
MSFT-TSM	0.3412	0.3874	0.4412	0.5124
GOOGL-AMD	0.3847	0.4241	0.4987	0.5641
GOOGL-TSM	0.3241	0.3712	0.4312	0.5012
AMD-TSM	0.2587	0.2987	0.3541	0.4312

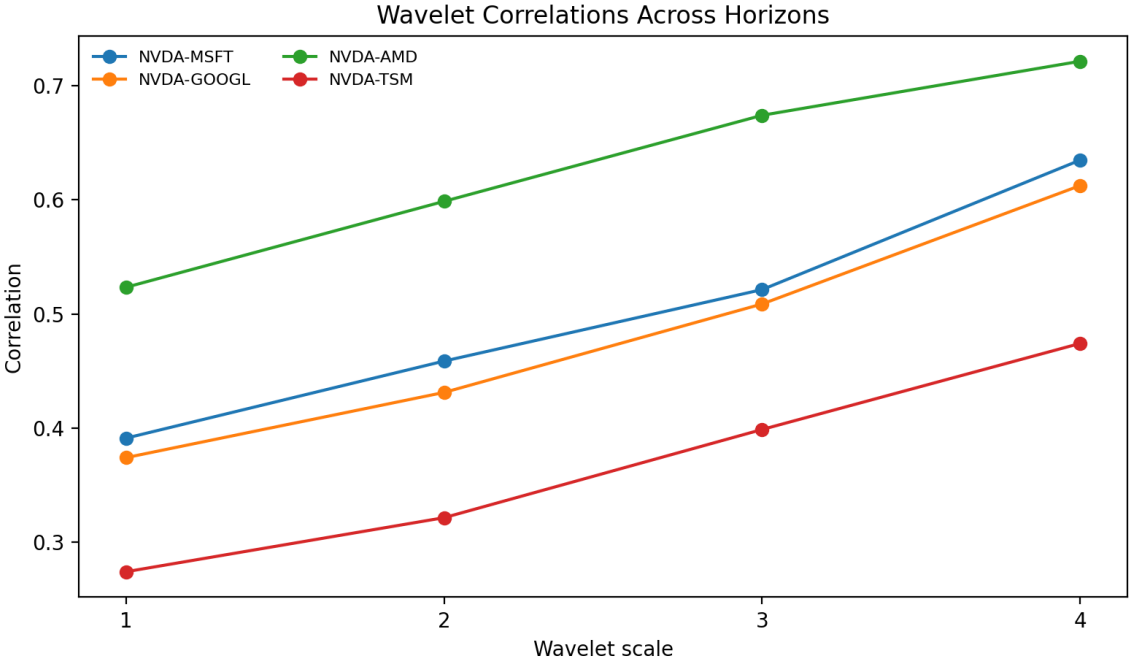
Appendix D. Result Chart Gallery



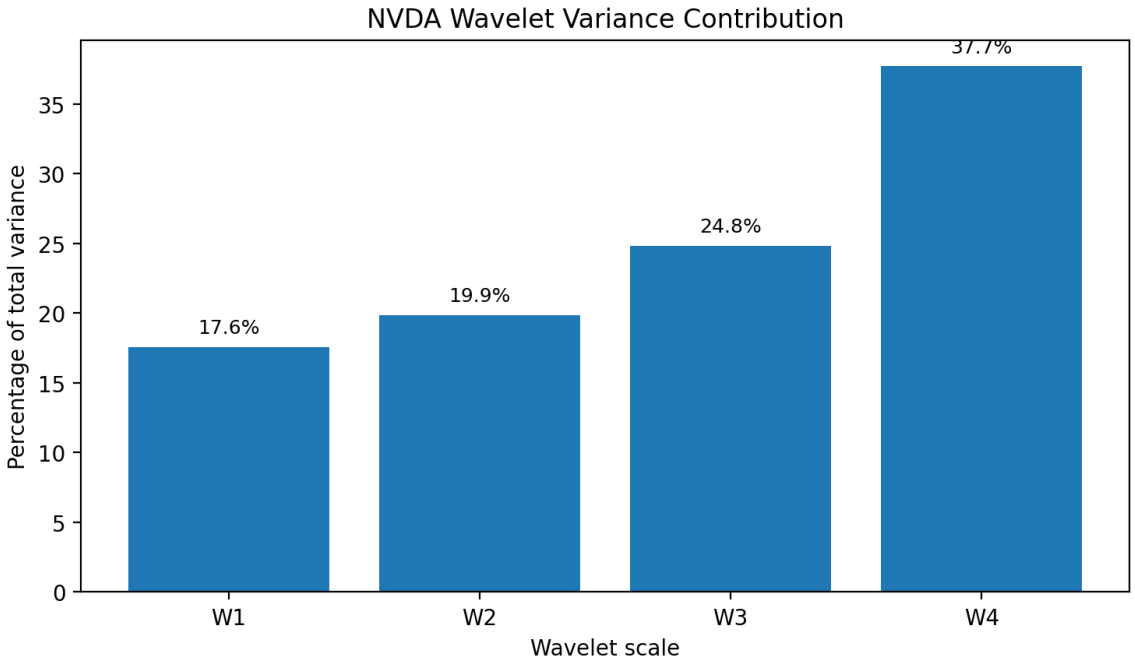
Volatility comparison.



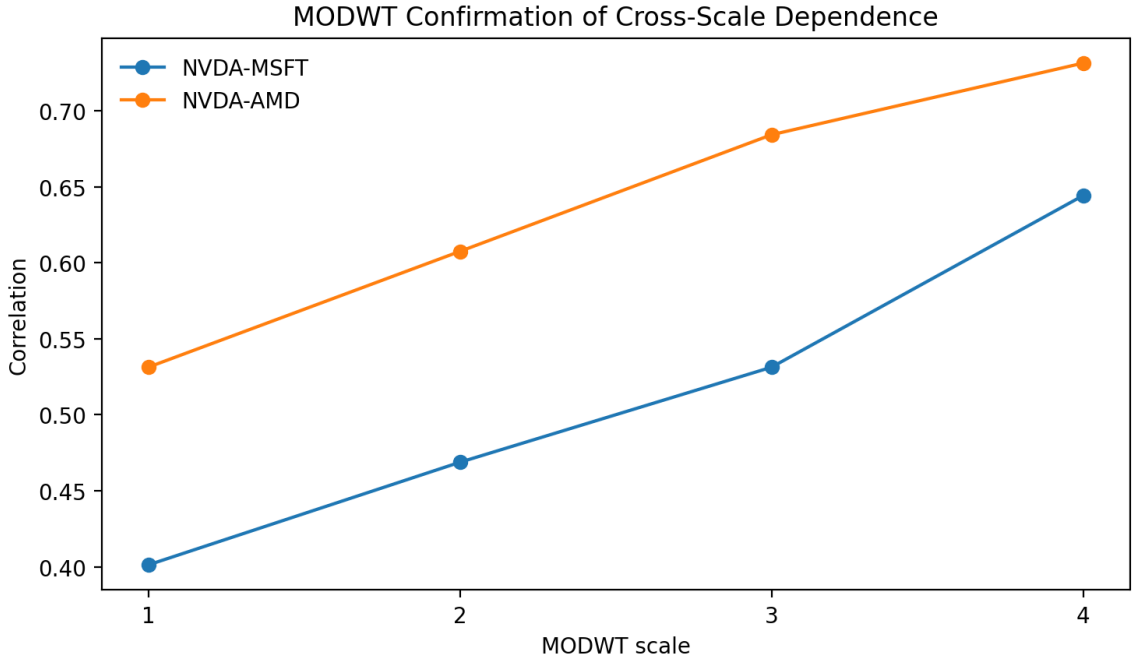
Correlation matrix.



Wavelet correlations.



NVDA variance decomposition.



MODWT confirmation.

Appendix E. R Workflow Summary

- 1. Load daily price data for the selected stocks.
- 2. Compute logarithmic returns using first differences of log prices.
- 3. Run descriptive statistics to identify the return distribution features.
- 4. Apply ADF, PP, and KPSS tests to confirm stationarity.
- 5. Check serial dependence and volatility clustering using Ljung-Box and ARCH LM tests.

- 6. Estimate DWT and MODWT wavelet decompositions to measure dependence across scales.
- 7. Select the VAR lag length using information criteria.
- 8. Estimate the VAR model and interpret the short-run spillovers.

Appendix F. Key Takeaways at a Glance

- AMD and NVIDIA are the most volatile names in the sample, which means they carry the most short-term risk.
- Microsoft and Alphabet show the strongest linear correlation, which reflects the platform-and-cloud side of the AI story.
- All return series are stationary, so the VAR framework is appropriate for the return data.
- ARCH effects are significant across all five stocks, which confirms volatility clustering.
- Wavelet results show that dependence becomes stronger as the horizon widens.
- MODWT confirms the same multiscale pattern and strengthens the interpretation.
- In the VAR system, AMD has a noticeable lagged influence on NVIDIA.
- TSMC is still connected to the group, but it remains the least tightly linked stock in the sample.

Appendix G. Extra Reading Notes

- The AI-tech cluster behaves more like a network than a simple basket of stocks.
- Short-run noise is present, but the more persistent cross-horizon dependence is the more important finding.
- The relationship between NVIDIA and AMD is strong enough to matter for both hedging and portfolio concentration.
- TSMC remains useful as a partial diversifier because it is connected, but not as tightly linked as the most correlated pairs.